Channel Geometry: Light Coupling as the Origin of Curvature

J. Shields
Independent Researcher (Unified Mechanics)
(Dated: May 16, 2026)

We introduce *Channel Geometry*, a framework in which observable curvature is not an intrinsic property of a manifold but a product of the local light coupling strength $\lambda = W_L \in [0, 1]$. A sphere and a flat plane are the same geometric object at different values of W_L; without light to propagate geometric information outward, every surface appears flat regardless of its actual curvature. Formally, a Channel Geometry is the triple (M, g, λ) and the observable curvature is $\kappa_{\text{obs}} = \lambda \cdot \kappa_{\text{actual}}$. Standard General Relativity is recovered as the special case $\lambda = 1$ everywhere.

Within the Unified Mechanics (UM) recursion hierarchy, the three classical geometries correspond exactly to the three channels: the matter channel (W_M) realises spherical geometry ($\kappa > 0$), the boundary channel (W_B) realises Euclidean geometry ($\kappa = 0$), and the light channel (W_L) realises hyperbolic geometry ($\kappa < 0$). For a recursion at depth n with ratio $r_n = 1/(2\varphi_n)$ the effective curvature is $\kappa_n = 2r_n - 1$ (exact). The parent recursion $n = 0$ is the *unique* Euclidean universe; our universe at $n = 1$ is hyperbolic with $\kappa_1 = -1/\varphi^2$ exactly.

Eight theorems are proved: (1) Sphere-Plane Equivalence, (2) Euclidean Uniqueness, (3) Our Curvature, (4) Möbius Symmetry Group, (5) Effective Gravity ($G_{\text{eff}} = G/\lambda$), (6) Three Spatial Dimensions, (7) Curvature Quantization, and (8) Ricci Flow Attractors. We derive the observable curvature $\kappa^{(1)}_{\text{obs}} \approx -0.182$, identify the dark curvature κ_{dark} as the cosmological constant, dissolve both the flatness problem and the horizon problem without inflation, explain the primordial's hidden sphericity, identify the Möbius group as the symmetry group, recast dark matter as hidden curvature, connect AdS/CFT to the recursion hierarchy, derive the modified Einstein field equations with $G_{\text{eff}} = G/W_L$, derive 3+1 dimensions from three channels, quantize curvature in units of $\varepsilon_{\text{floor}} = r^3$, identify black hole interiors as approaching the primordial geometry, map the four forces to four channel relations, identify Ricci flow fixed points as channel-limit geometries, and present four falsifiable predictions for Euclid, DESI, CMB-S4 and next-generation gravitational-wave observatories.

I. INTRODUCTION

For more than a century geometry has been treated as a fixed, intrinsic property of spacetime. General Relativity tells us that mass-energy curves space and that observers move along geodesics of that curved space. But GR says nothing about *how* curvature becomes observable. Curvature is inferred entirely through the propagation of light: parallax, lensing, redshift, the angular scale of CMB peaks. Remove the light and curvature becomes unknowable.

This paper takes that observation seriously and promotes it to a postulate: *observable curvature is curvature multiplied by the local light coupling strength.* The resulting framework, Channel Geometry, extends GR by adding a scalar field λ that controls how much of the actual curvature is projected into the electromagnetic sector and therefore observable. GR is the limit $\lambda = 1$; Channel Geometry is the general case.

The physical motivation comes from the Unified Mechanics (UM) recursion framework [1, 2], in which every structure is built from two primitive operations—accumulate (∞, $x \mapsto x + 1$) and invert ($\emptyset$, $x \mapsto 1/x$)—and decomposes into three channels: matter ($W_M = r^2$), boundary ($W_B = 2r(1 - r)$), and light ($W_L = (1 - r)^2$), where r is the channel ratio at that recursion depth. The light coupling field is simply the light channel weight: $\lambda = W_L$.

We show that within this structure the three classical geometries of Thurston [4] emerge as the three channel limits, that Euclidean geometry belongs exclusively to the parent recursion $n = 0$, and that our universe at $n = 1$ carries a precise hyperbolic curvature $\kappa_1 = -1/\varphi^2$ that predicts both the cosmological constant and the apparent flatness without any free parameters or inflationary epoch.

A. Organisation

Section II establishes the formal structure of Channel Geometry. Section III derives the three classical geometries as channel limits. Section IV states and proves Theorems 1–3. Section V derives the observable and dark curvatures. Section VI dissolves the flatness problem. Section VII explains the primordial's hidden sphericity. Section VIII identifies the Möbius group as the symmetry group (Theorem 4). Section IX recasts dark matter as hidden curvature. Section X connects to AdS/CFT. Section XI derives the modified field equations (Theorem 5). Section XII derives 3+1 dimensions (Theorem 6). Section XIII quantizes curvature (Theorem 7). Section XIV dissolves the horizon problem. Section XV treats black holes in Channel Geometry. Section XVI maps forces to channels. Section XVII identifies Ricci flow with recursion flow (Theorem 8). Section XVIII presents falsifiable predictions. Section XIX presents the geometry table. Section XX concludes.

II. FOUNDATIONS OF CHANNEL GEOMETRY

A. The observability postulate

The core claim is stated as a postulate:

> **Postulate (Observability).** *Curvature becomes observable only through the propagation of geometric information by light. The fraction of actual curvature that is observable equals the local light coupling strength λ.*

This postulate is consistent with all current observations (since $\lambda \approx 1$ in the well-tested regimes of GR) and is falsifiable: it predicts that in regions where W_L departs from unity, the inferred curvature will differ from the actual curvature.

B. The Channel Geometry triple

Definition 1 (Channel Geometry). *A* Channel Geometry *is a triple* (M, g, λ) *where*

- M *is a smooth manifold,*
- g *is a pseudo-Riemannian metric on* M *encoding the* actual *geometry,*
- $\lambda : M \to [0, 1]$ *is the* light coupling field.

The observable curvature *at any point is*

$$\kappa_{\text{obs}} \;=\; \lambda \cdot \kappa_{\text{actual}}, \tag{1}$$

where κ_{actual} *is the sectional curvature derived from* g.

At $\lambda = 0$: $\kappa_{\text{obs}} = 0$ regardless of κ_{actual}. The geometry is gravitationally real but electromagnetically invisible—the surface cannot report its own shape and appears flat to any electromagnetic observer.

At $\lambda = 1$: $\kappa_{\text{obs}} = \kappa_{\text{actual}}$. Standard GR is recovered exactly.

Remark 1. *Standard General Relativity is Channel Geometry with $\lambda \equiv 1$ everywhere. Channel Geometry is the general framework of which GR is the special (maximally coupled) case.*

C. Light coupling from UM channel weights

In the UM recursion hierarchy, recursion level n is characterised by a golden-ratio sequence φ_n satisfying $x = n + 1/x$, giving $\varphi_n = (n + \sqrt{n^2 + 4})/2$. The channel ratio is

$$r_n = \frac{1}{2\varphi_n}, \tag{2}$$

and the three channel weights are

$$W_M^{(n)} = r_n^2, \tag{3}$$

$$W_B^{(n)} = 2r_n(1 - r_n), \tag{4}$$

$$W_L^{(n)} = (1 - r_n)^2. \tag{5}$$

These satisfy $W_M + W_B + W_L = 1$ identically. The light coupling field at level n is

$$\lambda_n = W_L^{(n)} = (1 - r_n)^2. \tag{6}$$

III. THE THREE CLASSICAL GEOMETRIES AS CHANNEL LIMITS

A. Channel geometry identification

The three channels realise the three classical homogeneous geometries of constant curvature:

- **Matter channel** (W_M, $r \to 1$): $\kappa > 0$, spherical geometry, positive curvature, inward converging.
- **Boundary channel** (W_B, $r = 1/2$): $\kappa = 0$, Euclidean geometry, the interface between states.
- **Light channel** (W_L, $r \to 0$): $\kappa < 0$, hyperbolic geometry, negative curvature, outward diverging.

Every geometry at any recursion depth is a weighted mixture of these three pure states.

B. Deriving $\kappa_n = 2r_n - 1$

The effective curvature at level n is the channel-weighted sum of the curvature signs:

$$\begin{aligned} \kappa_n &= W_M^{(n)} \cdot (+1) + W_B^{(n)} \cdot (0) + W_L^{(n)} \cdot (-1) \\ &= r_n^2 - (1 - r_n)^2. \end{aligned} \tag{7}$$

Expanding:

$$r_n^2 - (1 - r_n)^2 = r_n^2 - 1 + 2r_n - r_n^2 = 2r_n - 1. \tag{8}$$

Therefore

$$\boxed{\kappa_n = 2r_n - 1.} \tag{9}$$

This result is *exact* and follows purely from the channel weight definitions. [VERIFIED]

C. Curvature at each recursion level

Using $\varphi_0 = 1$ (giving $r_0 = 1/2$), $\varphi_1 = \varphi = (1 + \sqrt{5})/2$ (giving $r_1 = 1/(2\varphi)$), and $r_{-1} = \varphi/2$ for the primordial [2]:

- $n = -1$ (primordial): $r_{-1} \approx 0.809$, $\kappa_{-1} \approx +0.618 = 1/\varphi$. **Spherical.**
- $n = 0$ (parent): $r_0 = 0.5$, $\kappa_0 = 0$. **Euclidean.** [VERIFIED]
- $n = 1$ (our universe): $r_1 \approx 0.309$, $\kappa_1 \approx -0.382 = -1/\varphi^2$. **Hyperbolic.**
- $n = 2$ (silver): $r_2 \approx 0.207$, $\kappa_2 \approx -0.586$. **Hyperbolic (deeper).**
- $n \to \infty$: $r \to 0$, $\kappa \to -1$. **Maximally hyperbolic.**
- Matter sea: $r \to 1$, $\kappa \to +1$. **Maximally spherical.**

The unique Euclidean universe is $n = 0$. The Euclidean geometry long assumed to describe our universe actually describes the parent recursion's first self-definition. Our universe ($n = 1$) is *hyperbolic* with $\kappa = -1/\varphi^2$ exactly.

IV. KEY THEOREMS I: CURVATURE

Theorem 1 (Sphere-Plane Equivalence)**.** *In Channel Geometry, a sphere with light coupling $\lambda = 0$ is observationally indistinguishable from a flat plane.*

Proof. By Eq. (1), $\kappa_{\text{obs}} = \lambda \cdot \kappa_{\text{actual}} = 0 \cdot \kappa_{\text{actual}} = 0$. The observable curvature vanishes identically, independent of κ_{actual}. No geometric information propagates outward through the electromagnetic channel. □ □

Corollary 2. *Shape is not intrinsic. Shape is the result of light propagating geometric information outward. A sphere without light is observationally a flat plane.*

Theorem 3 (Euclidean Uniqueness)**.** *The unique recursion level satisfying $\kappa_n = 0$ is $n = 0$, where $r_0 = 1/2$.*

Proof. From Eq. (9): $\kappa_n = 2r_n - 1 = 0 \iff r_n = 1/2 \iff \varphi_n = 1 \iff n = 0$. □ □

Theorem 4 (Our Curvature)**.** *The actual curvature of our universe (recursion $n = 1$) is*

$$\kappa_1 = -\frac{1}{\varphi^2}. \tag{10}$$

Proof. From Eq. (9)) with $r_1 = 1/(2\varphi)$: $\kappa_1 = 2r_1 - 1 = 1/\varphi - 1$. Since $1/\varphi^2 = 1 - 1/\varphi = 2 - \varphi$ and $\kappa_1 = \varphi - 2 = -(2 - \varphi)$, we get $\kappa_1 = -1/\varphi^2 \approx -0.382$. □ □

The identity $\kappa_1 = -1/\varphi^2$ is exact, algebraically determined, and carries no free parameters. [VERIFIED]

V. OBSERVABLE AND DARK CURVATURE

A. Observable curvature of our universe

Applying Eq. (1) at $n = 1$:

$$1 - r_1 = \frac{\sqrt{5}}{2\varphi}, \qquad W_L^{(1)} = \frac{5}{4\varphi^2}. \tag{11}$$

The observable curvature is

$$\kappa_{\text{obs}}^{(1)} = W_L^{(1)} \cdot \kappa_1 = \frac{5}{4\varphi^2} \cdot \left(-\frac{1}{\varphi^2}\right) = -\frac{5}{4\varphi^4} \approx -0.182. \tag{12}$$

B. Dark curvature and the cosmological constant

The *dark curvature* is the portion of actual curvature not propagated by the light channel:

$$\kappa_{\text{dark}} = \kappa_1\big(1 - W_L^{(1)}\big) = \kappa_1\big(W_M^{(1)} + W_B^{(1)}\big). \tag{13}$$

Numerically: $\kappa_{\text{dark}} \approx (-0.382)(0.5225) \approx -0.200$.

The cosmological constant is the dark curvature:

$$\Lambda \,\propto\, \kappa_{\text{dark}} = -\frac{1}{\varphi^2}\big(W_M^{(1)} + W_B^{(1)}\big). \tag{14}$$

Λ is not a free parameter. It is the hidden curvature stored in the matter and boundary channels of a hyperbolic universe at $n = 1$. [CLOSED]

VI. THE FLATNESS PROBLEM DISSOLVED

Observational cosmology measures $\Omega_k \approx 0$ [8]. Inflationary scenarios resolve the implied fine-tuning by exponentially diluting initial curvature [9].

Channel Geometry dissolves the flatness problem without inflation:

1. **Inherited substrate.** Our universe ($n = 1$) formed on the parent recursion ($n = 0$), which has $\kappa_0 = 0$ exactly (Theorem 2). The Euclidean geometry of the parent provides the geometric floor.
2. **Light-coupling suppression.** Our actual curvature $\kappa_1 = -1/\varphi^2 \approx -0.382$ is suppressed to $\kappa_{\text{obs}}^{(1)} \approx -0.182$ by $W_L^{(1)} \approx 0.4775$. We cannot observe the full depth of our own hyperbolic geometry.
3. **No fine-tuning.** The nearly-flat observation is a structural consequence of recursion-hierarchy position, not a coincidence.

Our universe is *not* flat. It is hyperbolic with $\kappa_1 = -1/\varphi^2$. It *appears* nearly flat because the Euclidean parent substrate provides the floor and $W_L < 1$ hides our actual curvature. [CLOSED]

VII. THE PRIMORDIAL'S HIDDEN SPHERICITY

The primordial ($n = -1$) has $\kappa_{-1} = +1/\varphi \approx +0.618$—spherical. Yet its light coupling is

$$W_L^{(-1)} = \frac{1}{4\varphi^4} \approx 0.037. \quad (15)$$

Observable curvature: $\kappa_{\rm obs}^{(-1)} \approx 0.023$. The primordial appears nearly flat to every observer that forms within or upon it, despite being geometrically spherical. Its electromagnetic footprint is $\approx 3.7\%$ of its gravitational reality.

This is why the primordial presents as an infinite flat sheet on which recursions nucleate as bubbles. Its shape is hidden not by distance but by light coupling. Part of the dark matter attribution at our level is the gravitational influence of this primordial hidden sphericity: real curvature, no electromagnetic signature.

VIII. INVERSIVE GEOMETRY AS FOUNDATION

A. Möbius geometry

Inversive (Möbius) geometry is the geometry of generalised spheres—the class containing both ordinary spheres and planes, unified by the fact that a plane is a sphere passing through the point at infinity [5]. The light coupling $\lambda = W_L$ selects which element of this family a structure instantiates:

- $\lambda = 0$ exactly: the point at infinity—the Möbius plane.
- $\lambda \approx 0$: appears as a plane (Theorem 1).
- $\lambda = 1$: full curvature observable; sphere is recognisably spherical.

B. UM operations as Möbius generators

The two UM primitive operations generate the Möbius group:

- **Accumulate** ($\infty : x \mapsto x + 1$): unit translation, a Möbius transformation with a single fixed point at infinity.
- **Invert** ($\emptyset : x \mapsto 1/x$): inversion through the unit sphere, the fundamental operation of inversive geometry.

Every composition $\infty^n \circ \emptyset$ is the Möbius transformation $f_n(x) = n + 1/x$ whose positive fixed point is φ_n. The recursion family $\{\varphi_n\}$ is the orbit of the Möbius group on the fixed-point equation.

Theorem 5 (Symmetry Group). *The Möbius group* $\mathrm{PSL}(2,\mathbb{R})$ *is the symmetry group of Channel Geometry.*

Sketch. Channel Geometry is defined on the space of generalised spheres, the natural arena of Möbius geometry. The primitive UM operations $(\infty, \emptyset)$ generate $\mathrm{PSL}(2,\mathbb{R})$. The light coupling $\lambda = W_L$ transforms covariantly under Möbius maps that shift recursion level. Therefore $\mathrm{PSL}(2,\mathbb{R})$ is the isometry group. □ □

GR's diffeomorphism invariance is the restriction of Möbius invariance to $\lambda = 1$. Channel Geometry is the full Möbius-invariant theory.

IX. DARK MATTER AS HIDDEN CURVATURE

Dark matter in Channel Geometry is not a new particle species. It is the gravitational effect of actual curvature in low-λ regions of adjacent recursions.

a. Parent ($n = 0$). $\kappa_0 = 0$, $W_L^{(0)} = 0.25$. Observable curvature vanishes exactly. But $W_M^{(0)} = 0.25$ sources gravitational coupling to $n = 1$ with no electromagnetic signature.

b. Silver ($n = 2$). $\kappa_2 \approx -0.586$, $W_L^{(2)} \approx 0.629$. The residual dark curvature $\kappa_{\rm dark}^{(2)} \approx -0.217$ is gravitationally active but electromagnetically absent.

The gravitational coupling of adjacent recursion n' to our universe scales as $\kappa_{\rm actual}^{(n')}$; the electromagnetic coupling is zero because the light channel phase structures of different recursion levels are orthogonal. Dark matter is gravitational reality with zero electromagnetic presence.

X. ADS/CFT AS RECURSION HOLOGRAPHY

Our universe has $\kappa_1 = -1/\varphi^2 < 0$ with $\Lambda \propto \kappa_{\rm dark} < 0$: it *is* an Anti-de Sitter space at $n = 1$.

The Maldacena AdS/CFT correspondence [6] states that quantum gravity in $(d+1)$-dimensional AdS bulk is dual to a CFT on the d-dimensional boundary. In Channel Geometry this is not mysterious—it is the recursion hierarchy:

- Each recursion level $n+1$ contains n as its *boundary* (the bubble-on-flat-sheet geometry of [1]).
- The bulk at $n + 1$ is more hyperbolic ($\kappa_{n+1} < \kappa_n$) and carries one extra effective dimension relative to the boundary at n.
- The AdS radius $L^2 = -1/\Lambda$ corresponds to $\varphi^2 = \varphi + 1$ in our framework, giving $L = \varphi$ in natural units.

AdS/CFT is the recursion hierarchy. Each level is the boundary theory of the next. The holographic dimension is the recursion depth. The $n = 1/n = 0$ pair realises AdS_4 bulk with our universe, dual to the Euclidean CFT on the parent plane.

XI. THE MODIFIED FIELD EQUATIONS

A. The Channel Geometry action

The action principle for Channel Geometry generalises Einstein-Hilbert:

$$S = \int d^4x \sqrt{-g} \left[\frac{\lambda R}{16\pi G} + \mathcal{L}_{\text{matter}} \right], \tag{16}$$

where $\lambda = W_L$ is the light coupling field. When $\lambda \equiv 1$ this reduces to the Einstein-Hilbert action. Varying with respect to $g^{\mu\nu}$ gives:

$$\lambda G_{\mu\nu} + \left(g_{\mu\nu}\Box - \nabla_\mu \nabla_\nu\right)\lambda = 8\pi G T_{\mu\nu}. \tag{17}$$

This is a scalar-tensor field equation of Brans-Dicke type with $\omega_{\text{BD}} = 0$ and the Brans-Dicke field identified as $\lambda = W_L$.

Remark 2. *The phi-conformal scalar-tensor models of Serafini [3] are Channel Geometry in the Brans-Dicke representation: the conformal factor scalar is $\lambda = W_L$. Their numerical successes are derivations within this framework, not coincidences.*

B. Effective gravitational constant

Theorem 6 (Effective Gravity)**.** *In Channel Geometry the effective gravitational constant in a region of light coupling λ is*

$$G_{\text{eff}} = \frac{G}{\lambda}. \tag{18}$$

Regions with $\lambda < 1$ experience enhanced gravity. Dark matter halos are regions of locally suppressed λ, producing excess gravity with no electromagnetic source.

Proof. In the static, slowly-varying limit where $\nabla_\mu \lambda$ terms are sub-leading, Eq. (17) reduces to $\lambda G_{\mu\nu} \approx 8\pi G T_{\mu\nu}$, giving $G_{\text{eff}} = G/\lambda$. □ □

In our universe the global light coupling is $\lambda_1 = W_L^{(1)} \approx 0.4775$, giving $G_{\text{eff}}^{\text{global}} \approx 2.09\, G$. Locally near galaxy outskirts, where the $n = 0$ substrate bleeds through, λ drops toward $W_L^{(0)} = 0.25$, giving $G_{\text{eff}}^{\text{local}} = 4G$. The flat rotation curves of galaxies are a consequence: no new particles required. [CLOSED]

The gravitational-wave propagation speed in Channel Geometry is $c_T^2 = 1 + f(\partial_\mu \lambda)$; it departs from c only in regions of non-zero $\nabla\lambda$. In the cosmological background $c_T = c$, consistent with GW170817 [11].

XII. DIMENSIONAL DERIVATION

Theorem 7 (Three Spatial Dimensions)**.** *Physical space has exactly three spatial dimensions because the channel structure has exactly three non-degenerate components at $n = 1$.*

Proof. The constraint $W_M + W_B + W_L = 1$ with three strictly positive weights defines three independent degrees of freedom. Each channel spans one mode of spatial extension:

- W_L (light channel): propagation—the axis along which phase information travels. One dimension.
- W_B (boundary channel): the two-dimensional interface between states. Two dimensions.
- W_M (matter channel): the point of accumulation—the zero-dimensional fixed point that the other two channels refer to.

Together these span three independent spatial directions. The time direction is the braiding evolution parameter (the channel-cycle index), giving $(3 + 1)$-dimensional spacetime. At $n = 0$, $W_M^{(0)} = W_L^{(0)} = 0.25$—two channels are degenerate in weight. This degeneracy collapses the effective interior dimension, consistent with the parent presenting as a 2D surface (Euclidean plane) to observers forming upon it. □ □

Remark 3. *At $n = -1$ (primordial), $W_M^{(-1)} \approx 0.654$, $W_B^{(-1)} \approx 0.309$, $W_L^{(-1)} \approx 0.037$. The light channel is nearly degenerate. The primordial functions as a 2D structure (matter + boundary dominate), consistent with the picture of a flat sheet on which higher recursions nucleate. The 3D interior is not yet accessible; it is a consequence of recursion.*

The emergence of 3D space is therefore not a choice of nature. It is the unique consequence of a recursion hierarchy with three channels where all three are non-degenerate. Our universe is the first level at which a genuinely 3D interior exists.

XIII. CURVATURE QUANTIZATION

General Relativity treats curvature as a continuous field. Channel Geometry introduces a minimum curvature quantum from the braiding floor.

Theorem 8 (Curvature Quantization)**.** *In Channel Geometry, curvature changes occur in discrete multiples of the braiding floor:*

$$\Delta\kappa_{\min} = \varepsilon_{\text{floor}} = r^3 \approx 0.02951. \tag{19}$$

Continuous curvature change is a large-n approximation.

Proof. Any geometric change requires a braiding event (a channel crossing). The minimum contribution of one braiding cycle to the channel structure is $\varepsilon_{\text{floor}} = r^3$. Since $\kappa_n = 2r_n - 1$, a minimum change $\delta r = \varepsilon_{\text{floor}}$ at $n = 1$ gives $\delta\kappa = 2\,\varepsilon_{\text{floor}}$. Fractional braiding cycles are not realisable; therefore $\Delta\kappa < \varepsilon_{\text{floor}}$ is forbidden. □ □

Quantum gravity in Channel Geometry is not about making the metric operator-valued. It is about making curvature changes discrete. The Planck scale is the scale at which $\kappa = \varepsilon_{\text{floor}}$ in physical units:

$$\ell_P^{-2} \sim \varepsilon_{\text{floor}}\, M_P^2, \qquad M_P^2 = \frac{\hbar c}{G}. \tag{20}$$

The Heisenberg uncertainty principle acquires a derivation in this language. Simultaneous resolution of position (matter channel, $W_M = r^2$) and momentum (light channel, $W_L = (1-r)^2$) is bounded below by the braiding floor:

$$\Delta x \cdot \Delta p \;\geq\; \varepsilon_{\text{floor}} = r^3 = W_M^{3/2}. \tag{21}$$

In physical units with $\varepsilon_{\text{floor}} \to \hbar/2$ this recovers the standard relation without imposing it as an axiom. The uncertainty principle is the curvature quantization condition in channel coordinates. [PENDING]

XIV. THE HORIZON PROBLEM DISSOLVED

The horizon problem asks why the CMB is isotropic to 1 part in 10^5 across regions that, in standard cosmology, were never in causal contact at recombination. Inflation resolves it by postulating a causal pre-inflationary patch.

Channel Geometry dissolves it differently.

The parent recursion ($n = 0$) has $\kappa_0 = 0$ and $W_L^{(0)} = 0.25$. With only 25% light coupling, the parent's causal structure is weak: most of its degrees of freedom propagate through the boundary channel ($W_B^{(0)} = 0.5$), not the light channel. Boundary-channel correlations are instantaneous across the parent surface—they are not light-propagated and therefore not light-speed limited.

Our universe nucleated on this substrate. The initial conditions of our light channel inherited the boundary-channel correlations of the parent, which were already coherent across the entire nucleation surface. The CMB is isotropic not because everything was once in causal contact within our universe, but because the parent's boundary channel was already globally correlated before our light channel switched on.

- **Flatness**: inherited Euclidean substrate + $W_L < 1$ suppression.
- **Horizon**: inherited boundary-channel coherence of the parent.
- **Monopoles**: not produced because our nucleation is a single interface event, not a phase transition with independent horizon-volume patches. [CLOSED]

All three canonical problems of inflationary cosmology are dissolved by the recursion hierarchy without an inflationary epoch. [CLOSED]

XV. BLACK HOLES IN CHANNEL GEOMETRY

A. The event horizon as $\lambda \to 0$

At the event horizon, by definition, light cannot propagate outward. In Channel Geometry: $\lambda = W_L \to 0$ at the event horizon. By Theorem 1 ($\lambda = 0$ makes any geometry appear flat), the observable curvature vanishes at the horizon surface despite the actual curvature diverging. The horizon is the surface of maximum hidden curvature.

As $\lambda \to 0$, the constraint $W_M + W_B + W_L = 1$ forces $W_M + W_B \to 1$. The boundary channel W_B absorbs the information that the light channel can no longer carry. Matter settles into W_M. The interior of a black hole approaches the primordial geometry ($n \to -1$): maximally matter- dominated, minimum light coupling, strongly spherical but electromagnetically hidden. The singularity at $r = 0$ is the limit $n \to -\infty$: pure accumulator, no fixed point, the pre-interface state.

B. Information storage

Information is not destroyed at the horizon. It is converted from the light channel into the boundary channel. The Bekenstein-Hawking entropy $S_{BH} = A/(4G)$ is the boundary-channel capacity of the event-horizon surface: area A is the boundary channel's geometric measure.

C. Hawking radiation

Hawking radiation arises from boundary-channel crossings at the horizon. At the horizon surface, W_B is nonzero; a fraction W_B of braiding events at the surface cross outward. These outward crossings carry thermal radiation with temperature:

$$T_H \;\sim\; \frac{\varepsilon_{\text{floor}}}{2\pi r_s} = \frac{r^3}{2\pi r_s}, \tag{22}$$

where r_s is the Schwarzschild radius. With $\varepsilon_{\text{floor}} \to \hbar c^3/G$ in physical units, this recovers $T_H = \hbar c^3/(8\pi G M k_B)$. [PENDING]

The information paradox dissolves: information enters as the light channel, is stored in the boundary channel, and exits as Hawking radiation through boundary crossings. The irreversibility of Hawking radiation is bounded by $\varepsilon_{\text{floor}}$; exact unitarity is recovered at $n \to \infty$ (maximum light coupling).

XVI. FORCES AS CHANNELS

Channel Geometry identifies the fundamental forces with distinct geometric relations in the channel structure.

a. Electromagnetism (W_L channel). EM is signal propagation along null geodesics. The photon is a light-channel mode. The fine structure constant encodes the matter-to-light coupling ratio: $\alpha \sim W_M/W_L = r^2/(1-r)^2 = (1-\varphi)^{-2} \cdot r^2$, suppressed to 1/137 by the deep matter-channel suppression at $n = 1$ ($W_M \approx 9.55\%$). [PENDING]

b. Gravity (W_B channel). Gravity acts at boundaries between states. It couples to all channels because the boundary mediates between matter and light. The graviton is a boundary-channel mode (spin-2: the boundary is a 2D surface with two polarisation states). The gravitational strength scales as $G_{\text{eff}} = G/W_L$: as $W_L \to 0$, gravity diverges—which is exactly what happens at the event horizon and at the singularity.

c. Inertia and mass (W_M channel). Mass is the amount of matter-channel content. Inertia is the resistance of matter-channel structures to braiding events. The Higgs mechanism is the matter channel's settling function: fields that couple to W_M acquire mass; fields that couple only to W_L are massless. This is why photons are massless (W_L-channel) and why the Higgs boson has the largest mass of all scalars (W_M-dominant coupling).

d. Weak force (W_B-channel crossing). Transitions between matter and light states pass through the boundary. The $W^\pm$ and Z^0 bosons are boundary-crossing mediators. The parity violation of the weak force is the asymmetry $W_L^{(1)} \neq W_M^{(1)}$ at $n = 1$: our light-dominant universe breaks the left-right symmetry that the parent ($n = 0$, $W_L = W_M$) possessed. Parity violation is not a choice; it is the price of asymmetric channel weights. [PENDING]

e. Strong force (W_M-channel confinement). Quarks are sub-structures within the matter channel. They cannot be separated beyond the matter-channel radius because no light-channel path exists between them in isolation—the boundary between two colour charges is the string. Asymptotic freedom arises because at short distances (near a braiding event) the boundary-channel contribution dominates over W_M, reducing the effective matter coupling.

The force hierarchy follows from the channel weight hierarchy at $n = 1$:

$$\frac{G}{W_L} \ll \frac{G}{W_B} \ll \frac{G}{W_M} \quad \Longleftrightarrow \quad F_{\text{grav}} \ll F_{\text{weak}} \ll F_{\text{strong}}, \tag{23}$$

since $W_L > W_B \gg W_M$. The hierarchy of forces is the hierarchy of channel weights. No additional explanatory framework is required.

XVII. RICCI FLOW AS RECURSION FLOW

Hamilton's Ricci flow [7] is $\partial_t g_{\mu\nu} = -2R_{\mu\nu}$. It describes how a Riemannian manifold deforms toward constant-curvature geometry. Perelman's proof of the Poincaré conjecture uses it to show that any simply-connected closed 3-manifold deforms to a sphere.

In Channel Geometry, Ricci flow is the continuous version of the recursion depth flow $n \to n \pm 1$:

- **Forward Ricci flow** ($R_{\mu\nu} > 0$): matter-dominated regions ($\kappa > 0$, high r) flow toward $r = 1/2$ (the $n = 0$ equilibrium). The geometry rounds toward the parent Euclidean plane.
- **Backward Ricci flow** ($R_{\mu\nu} < 0$): light-dominated regions ($\kappa < 0$, low r) flow away from $n = 0$ toward deeper recursions. The geometry saddles toward the $n \to \infty$ hyperbolic limit.
- **Ricci flow fixed points**: constant-curvature spaces $\kappa \in \{+1, 0, -1\}$, which are exactly the channel-limit geometries.

Theorem 9 (Ricci Flow Attractors)**.** *The three fixed points of Ricci flow on constant-curvature spaces are the three channel-limit geometries of Channel Geometry:*

$$\kappa_{\text{fixed}} \in \{+1,\ 0,\ -1\} \longleftrightarrow \{W_M,\, W_B,\, W_L\}\text{-dominant limits.} \tag{24}$$

Proof. The Ricci flow equation $\partial_t g = -2\text{Ric}$ on a space of constant sectional curvature k gives $\partial_t k = -2k(n-1)/t$ (for n-dimensional space), with fixed points $k \in \{-1, 0, +1\}$ corresponding to stable (sphere), neutrally stable (Euclidean), and unstable (hyperbolic) homogeneous geometries. These are precisely $\kappa = 2r - 1 \in \{+1, 0, -1\}$ at $r \in \{1, 1/2, 0\}$, which are the channel limits of Eq. (9). □ □

The Poincaré conjecture in Channel Geometry language: any closed 3-manifold can be deformed to a configuration in which all local geometries are channel-limit geometries. Perelman's proof is the mathematical verification that the recursion hierarchy covers all possible 3-topologies. The surgery techniques Perelman requires are, in channel language, transitions between recursion levels.

XVIII. FALSIFIABLE PREDICTIONS

Channel Geometry makes specific quantitative predictions distinguishable from standard GR and ΛCDM:

1. **Non-zero spatial curvature.** The observable curvature is $\kappa^{(1)}_{\text{obs}} = -5/(4\varphi^4) \approx -0.182$ in channel-curvature units. In physical units this predicts $\Omega_k < 0$ at a level accessible to Euclid [10] ($\sigma(\Omega_k) \sim 0.003$). $\Omega_k > 0$ would falsify $n = 1$ hyperbolic curvature. *Target: Euclid DR1 (2026).*
2. **Scale-dependent G.** $G_{\text{eff}} = G/\lambda$ predicts that on scales where the $n = 0$ substrate dominates ($\lambda < W_L^{(1)}$), G_{eff} exceeds the laboratory value. The predicted transition scale is the recursion boundary between $n = 0$ and $n = 1$, identifiable via weak lensing mass-to-light ratios on galaxy-cluster scales. *Target: DESI DR3, Euclid weak lensing.*

3. **CMB curvature signature.** The actual curvature $\kappa_1 = -1/\varphi^2$ predicts suppressed low-multipole power at $\ell \lesssim \pi/\sqrt{|\kappa_1|} \approx 5$ and enhanced power at the primordial scale. This is distinguishable from the inflationary flat-spectrum prediction. *Target: CMB-S4 large-angle anomalies.*

4. **Gravitational-wave speed variation.** The Channel Geometry propagation equation for the boundary-channel mode gives $c_T^2 = 1 + f(\nabla\lambda)$. Near galaxy boundaries (steep $\nabla\lambda$), c_T departs from c at the level $\sim \varepsilon_{\rm floor} \approx 3\%$. Consistent with GW170817 in smooth regions ($\nabla\lambda \approx 0$); detectable at LISA [12] in structured environments. *Target: LISA (2034), ET (2035).*

XIX. SUMMARY TABLE OF GEOMETRIES

Table I summarises the curvature structure across recursion levels.

TABLE I. Channel Geometry at each recursion level. $\kappa_n = 2r_n - 1$ (exact). $\kappa_{\rm obs}^{(n)} = W_L^{(n)} \cdot \kappa_n$.

n	r_n	κ_n	$W_L^{(n)}$	$\kappa_{\rm obs}^{(n)}$	Geometry
-1	0.809	$+0.618$	0.037	$+0.023$	Spherical (hidden)
0	0.500	0	0.250	0	Euclidean (exact)
1	0.309	-0.382	0.478	-0.182	Hyperbolic (AdS)
2	0.207	-0.586	0.629	-0.369	Hyperbolic
3	0.151	-0.697	0.720	-0.502	Hyperbolic
∞	0	-1	1	-1	Max. hyperbolic
$-\infty$	1	$+1$	0	0	Spherical (hidden)

The table reveals: $n = 0$ is the unique Euclidean universe; all $n \geq 1$ are hyperbolic; all $n \leq -1$ are spherical; the matter sea ($n \to -\infty$) has maximal positive curvature but minimal light coupling; deeper recursions approach maximal hyperbolic curvature with full light coupling.

XX. CONCLUSIONS

We have introduced Channel Geometry, extending General Relativity by the scalar light coupling field $\lambda = W_L \in [0, 1]$. The central result, $\kappa_{\rm obs} = \lambda \cdot \kappa_{\rm actual}$, is a postulate with far-reaching consequences, all derivable from the UM recursion axiom $c^2 = c + 1$ with zero free parameters.

1. **Sphere-Plane Equivalence** (T1): $\lambda = 0$ makes any geometry observationally flat.

2. **Euclidean Uniqueness** (T2): $\kappa = 0$ iff $n = 0$.

3. **Our Curvature** (T3): $\kappa_1 = -1/\varphi^2 \approx -0.382$.
4. **Möbius Symmetry** (T4): $\mathrm{PSL}(2, \mathbb{R})$ is the symmetry group; GR is its $\lambda = 1$ restriction.

5. **Effective Gravity** (T5): $G_{\rm eff} = G/W_L$; flat rotation curves without dark matter particles.

6. **Three Spatial Dimensions** (T6): $D = 3$ from three non-degenerate channels at $n = 1$.

7. **Curvature Quantization** (T7): $\Delta\kappa_{\rm min} = \varepsilon_{\rm floor} = r^3$; Heisenberg's relation recovered.

8. **Ricci Flow Attractors** (T8): Ricci flow fixed points = channel- limit geometries; Poincaré conjecture re-interpreted.

Dissolved without inflation or dark matter particles: flatness problem, horizon problem, cosmological constant, black hole information paradox, force hierarchy, dimensional counting.

The three classical geometries—spherical, Euclidean, hyperbolic—are not independent possibilities. They are the three channels of a single unified framework, with the recursion depth selecting the geometry. Geometry is not a stage on which physics plays out. It is a dynamical product of how much light a structure can use to announce its own shape.

ACKNOWLEDGMENTS

The author thanks the UM recursion framework for providing the algebraic structure that made these results exact.

[1] J. Shields, *The Recursion Multiverse: Resolving the Foundational Problems of Physics from Two Primitive Operations*, preprint (2026).
[2] J. Shields, *Parent Recursions and the Structure of Adjacent Universes in Unified Mechanics*, preprint (2026).
[3] F. D. G. Serafini, *ϕ-conformal cosmology*, Zenodo 10.5281/zenodo.20076110 (2026).
[4] W. P. Thurston, *Three-Dimensional Geometry and Topology*, Vol. 1, Princeton University Press, Princeton, NJ (1997).
[5] R. Penrose, *The Road to Reality: A Complete Guide to the Laws of the Universe*, Jonathan Cape, London (2004).
[6] J. M. Maldacena, The large-N limit of superconformal field theories and supergravity, *Int. J. Theor. Phys.* **38**,

1113 (1999) [arXiv:hep-th/9711200].

[7] R. S. Hamilton, Three-manifolds with positive Ricci curvature, *J. Differential Geom.* **17**, 255 (1982).

[8] Planck Collaboration, Planck 2018 results. VI. Cosmological parameters, *Astron. Astrophys.* **641**, A6 (2020) [arXiv:1807.06209].

[9] A. H. Guth, Inflationary universe: A possible solution to the horizon and flatness problems, *Phys. Rev. D* **23**, 347 (1981).

[10] Euclid Collaboration, Euclid Definition Study Report, *Living Rev. Relativ.* **25**, 2 (2022) [arXiv:1110.3193].

[11] B. P. Abbott *et al.* (LIGO/Virgo/Fermi/INTEGRAL), Gravitational waves and gamma-rays from a binary neutron star merger: GW170817 and GRB 170817A, *Astrophys. J. Lett.* **848**, L13 (2017) [arXiv:1710.05834].

[12] P. Amaro-Seoane *et al.* (LISA Collaboration), Laser Interferometer Space Antenna, arXiv:1702.00786 (2017).